

```
In [47]: import matplotlib.pyplot as plt
import pandas as pd
import numpy as np
import warnings
warnings.filterwarnings('ignore')
import seaborn as sns

%matplotlib inline
plt.rcParams['figure.figsize'] = 8,4
```

```
In [48]: df = pd.read_csv(r'C:\Users\User\Desktop\GDP analysis.csv')
df
```

```
Out[48]:
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

```
In [49]: len(df)
```

```
Out[49]: 195
```

```
In [50]: df.shape
```

```
Out[50]: (195, 5)
```

```
In [51]: df.columns
```

```
Out[51]: Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',
               'IncomeGroup'],
              dtype='object')
```

```
In [52]: df_catg = df[['CountryName', 'CountryCode',      #catg or text colounm of data fr
                    'IncomeGroup']]
df_catg
```

```
Out[52]:
```

	CountryName	CountryCode	IncomeGroup
0	Aruba	ABW	High income
1	Afghanistan	AFG	Low income
2	Angola	AGO	Upper middle income
3	Albania	ALB	Upper middle income
4	United Arab Emirates	ARE	High income
...	...	...	...
190	Yemen, Rep.	YEM	Lower middle income
191	South Africa	ZAF	Upper middle income
192	Congo, Dem. Rep.	COD	Low income
193	Zambia	ZMB	Lower middle income
194	Zimbabwe	ZWE	Low income

195 rows × 3 columns

```
In [53]: df_num = df[['CountryCode', 'BirthRate']] #numrical part of Data frame
df_num
```

```
Out[53]:
```

	CountryCode	BirthRate
0	ABW	10.244
1	AFG	35.253
2	AGO	45.985
3	ALB	12.877
4	ARE	11.044
...	...	...
190	YEM	32.947
191	ZAF	20.850
192	COD	42.394
193	ZMB	40.471
194	ZWE	35.715

195 rows × 2 columns

```
In [54]: df.head()
```

Out[54]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income

In [55]: `df.tail()`

Out[55]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

In [56]: `df[['CountryCode', 'BirthRate']].head()`

Out[56]:

	CountryCode	BirthRate
0	ABW	10.244
1	AFG	35.253
2	AGO	45.985
3	ALB	12.877
4	ARE	11.044

In [57]: `df_catg.head()`

Out[57]:

	CountryName	CountryCode	IncomeGroup
0	Aruba	ABW	High income
1	Afghanistan	AFG	Low income
2	Angola	AGO	Upper middle income
3	Albania	ALB	Upper middle income
4	United Arab Emirates	ARE	High income

In [58]: `df_num.head()`

Out[58]:

	CountryCode	BirthRate
0	ABW	10.244
1	AFG	35.253
2	AGO	45.985
3	ALB	12.877
4	ARE	11.044

In [59]: `df[['BirthRate']]`

Out[59]:

	BirthRate
0	10.244
1	35.253
2	45.985
3	12.877
4	11.044
...	...
190	32.947
191	20.850
192	42.394
193	40.471
194	35.715

195 rows × 1 columns

In [60]: `df[['CountryName', 'CountryCode', 'BirthRate']]`

Out[60]:

	CountryName	CountryCode	BirthRate
0	Aruba	ABW	10.244
1	Afghanistan	AFG	35.253
2	Angola	AGO	45.985
3	Albania	ALB	12.877
4	United Arab Emirates	ARE	11.044
...	...	...	...
190	Yemen, Rep.	YEM	32.947
191	South Africa	ZAF	20.850
192	Congo, Dem. Rep.	COD	42.394
193	Zambia	ZMB	40.471
194	Zimbabwe	ZWE	35.715

195 rows × 3 columns

In [61]: `df[['CountryName', 'CountryCode', 'BirthRate']].tail()`

Out[61]:

	CountryName	CountryCode	BirthRate
190	Yemen, Rep.	YEM	32.947
191	South Africa	ZAF	20.850
192	Congo, Dem. Rep.	COD	42.394
193	Zambia	ZMB	40.471
194	Zimbabwe	ZWE	35.715

In [62]: `df[:]`

Out[62]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

In [63]: `df[:11]`

Out[63]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9000	High income
1	Afghanistan	AFG	35.253	5.9000	Low income
2	Angola	AGO	45.985	19.1000	Upper middle income
3	Albania	ALB	12.877	57.2000	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0000	High income
5	Argentina	ARG	17.716	59.9000	High income
6	Armenia	ARM	13.308	41.9000	Lower middle income
7	Antigua and Barbuda	ATG	16.447	63.4000	High income
8	Australia	AUS	13.200	83.0000	High income
9	Austria	AUT	9.400	80.6188	High income
10	Azerbaijan	AZE	18.300	58.7000	Upper middle income

In [64]: `df[:, -1]`

Out[64]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
194	Zimbabwe	ZWE	35.715	18.5	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
191	South Africa	ZAF	20.850	46.5	Upper middle income
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
...	...	...	...	...	...
4	United Arab Emirates	ARE	11.044	88.0	High income
3	Albania	ALB	12.877	57.2	Upper middle income
2	Angola	AGO	45.985	19.1	Upper middle income
1	Afghanistan	AFG	35.253	5.9	Low income
0	Aruba	ABW	10.244	78.9	High income

195 rows × 5 columns

```
In [65]: df['mycac1'] = df.BirthRate*df.InternetUsers
df['mycac1']
```

```
Out[65]: 0      808.2516
1      207.9927
2      878.3135
3      736.5644
4      971.8720
...
190    658.9400
191    969.5250
192     93.2668
193    623.2534
194    660.7275
Name: mycac1, Length: 195, dtype: float64
```

In [66]: df

Out[66]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup	mycacl
0	Aruba	ABW	10.244	78.9	High income	808.2516
1	Afghanistan	AFG	35.253	5.9	Low income	207.9927
2	Angola	AGO	45.985	19.1	Upper middle income	878.3135
3	Albania	ALB	12.877	57.2	Upper middle income	736.5644
4	United Arab Emirates	ARE	11.044	88.0	High income	971.8720
...	...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income	658.9400
191	South Africa	ZAF	20.850	46.5	Upper middle income	969.5250
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income	93.2668
193	Zambia	ZMB	40.471	15.4	Lower middle income	623.2534
194	Zimbabwe	ZWE	35.715	18.5	Low income	660.7275

195 rows × 6 columns

In [67]: `df.describe()`

Out[67]:

	BirthRate	InternetUsers	mycacl
count	195.000000	195.000000	195.000000
mean	21.469928	42.076471	653.559009
std	10.605467	29.030788	351.553521
min	7.900000	0.900000	28.990400
25%	12.120500	14.520000	361.263300
50%	19.680000	41.000000	682.074300
75%	29.759500	66.225000	892.690170
max	49.661000	96.546800	1552.589500

In [68]: `df.describe().T`

Out[68]:

	count	mean	std	min	25%	50%	75%
BirthRate	195.0	21.469928	10.605467	7.9000	12.1205	19.6800	29.75950
InternetUsers	195.0	42.076471	29.030788	0.9000	14.5200	41.0000	66.22500
mycac1	195.0	653.559009	351.553521	28.9904	361.2633	682.0743	892.69017

In [69]:

```
df[['CountryName','BirthRate']][4:10]
```

Out[69]:

	CountryName	BirthRate
4	United Arab Emirates	11.044
5	Argentina	17.716
6	Armenia	13.308
7	Antigua and Barbuda	16.447
8	Australia	13.200
9	Austria	9.400

In [70]:

```
df[0:12][['mycac1','BirthRate']]
```

Out[70]:

	mycac1	BirthRate
0	808.25160	10.244
1	207.99270	35.253
2	878.31350	45.985
3	736.56440	12.877
4	971.87200	11.044
5	1061.18840	17.716
6	557.60520	13.308
7	1042.73980	16.447
8	1095.60000	13.200
9	757.81672	9.400
10	1074.21000	18.300
11	57.39630	44.151

In [71]:

```
df = df.drop('mycac1',axis = 1)
df
```

Out[71]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

In [72]: `len(df[df.IncomeGroup == 'Low income'])`

Out[72]: 30

In [73]: `df[df.IncomeGroup == 'High income']`

Out[73]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.90	High income
4	United Arab Emirates	ARE	11.044	88.00	High income
5	Argentina	ARG	17.716	59.90	High income
7	Antigua and Barbuda	ATG	16.447	63.40	High income
8	Australia	AUS	13.200	83.00	High income
...	...	...	...	...	...
174	Trinidad and Tobago	TTO	14.590	63.80	High income
180	Uruguay	URY	14.374	57.69	High income
181	United States	USA	12.500	84.20	High income
184	Venezuela, RB	VEN	19.842	54.90	High income
185	Virgin Islands (U.S.)	VIR	10.700	45.30	High income

67 rows × 5 columns

In [74]: `len(df[df.IncomeGroup == 'High income'])`

Out[74]: 67

In [75]: `df[df.BirthRate<=40]`

Out[75]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
5	Argentina	ARG	17.716	59.9	High income
...	...	...	...	...	...
188	West Bank and Gaza	PSE	30.394	46.6	Lower middle income
189	Samoa	WSM	26.172	15.3	Lower middle income
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

183 rows × 5 columns

In [76]: `df[(df.BirthRate>35) & (df.InternetUsers>10)]`

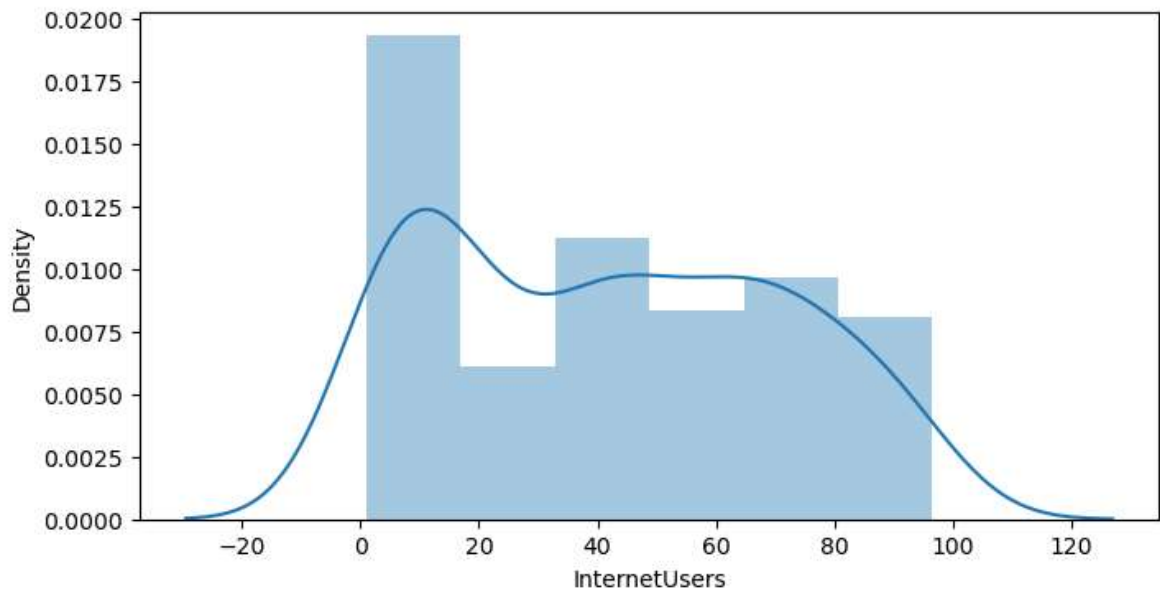
Out[76]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
2	Angola	AGO	45.985	19.1	Upper middle income
65	Gambia, The	GMB	42.525	14.0	Low income
67	Equatorial Guinea	GNQ	35.362	16.4	High income
91	Kenya	KEN	35.194	39.0	Lower middle income
128	Nigeria	NGA	40.045	38.0	Lower middle income
151	Senegal	SEN	38.533	13.1	Lower middle income
158	South Sudan	SSD	37.126	14.1	Low income
178	Uganda	UGA	43.474	16.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

visualization

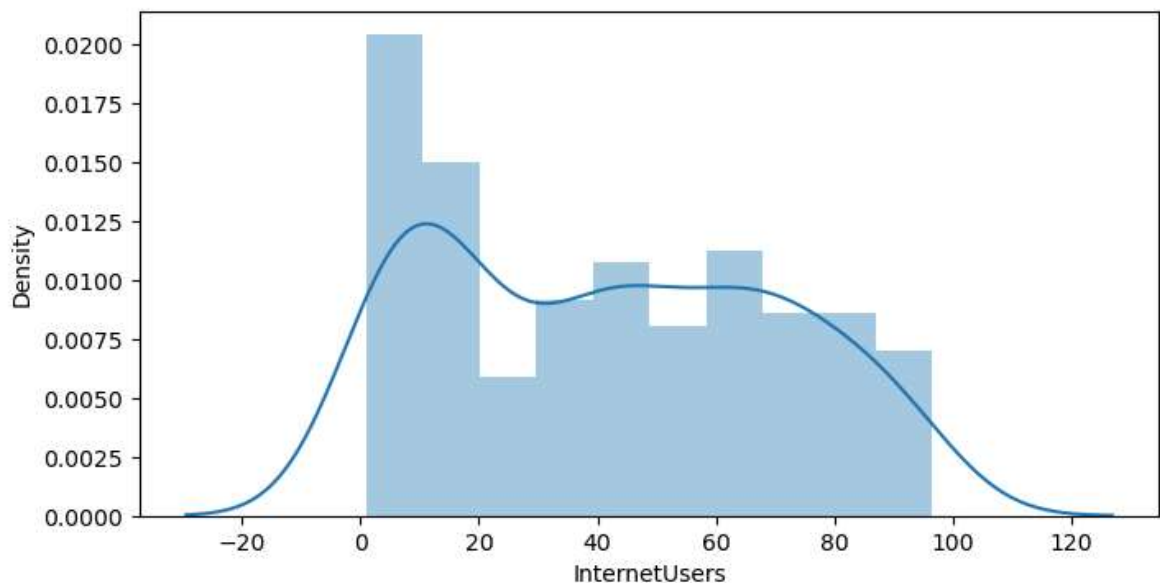
In [84]: `vis1 = sns.distplot(df['InternetUsers']) #univariate
vis1`

Out[84]: <Axes: xlabel='InternetUsers', ylabel='Density'>

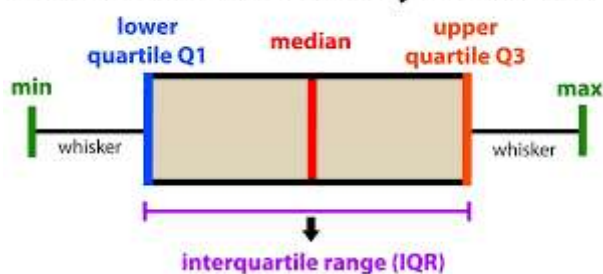


```
In [78]: vis1 = sns.distplot(df['InternetUsers'], bins = 10) #univariate
vis1
```

Out[78]: <Axes: xlabel='InternetUsers', ylabel='Density'>



introduction to data analysis: Box Plot



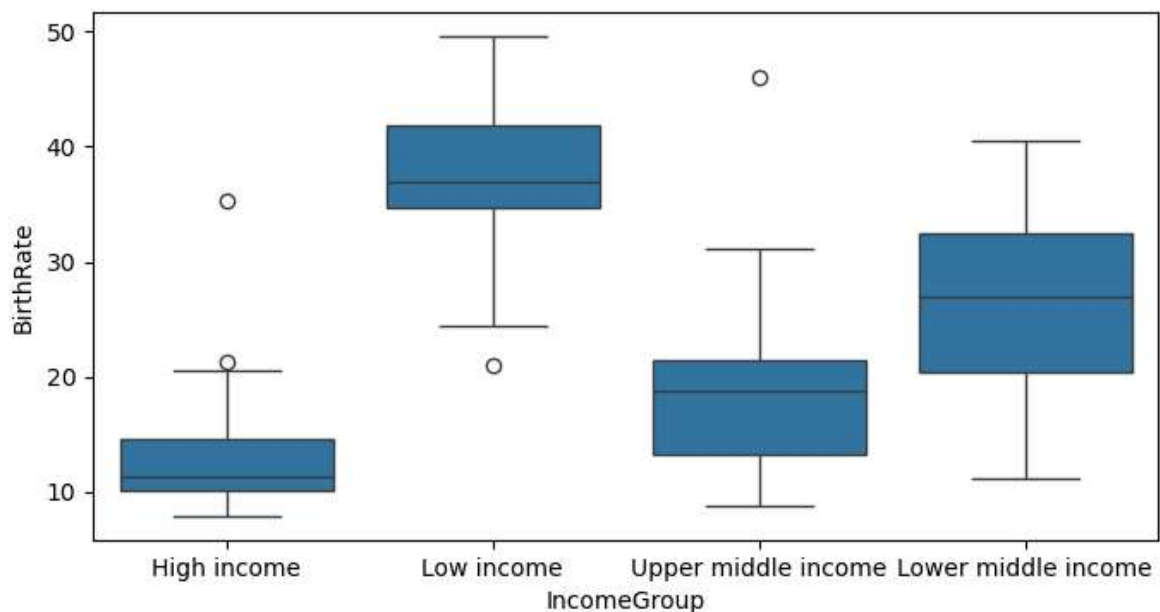
boxplot = a method for visualizing the distribution of quantitative data across

categories, showing key statistical summaries like median, quartiles, and outliers.

outliers = In Seaborn, an outlier in a boxplot is defined as any data point that lies outside the whiskers.

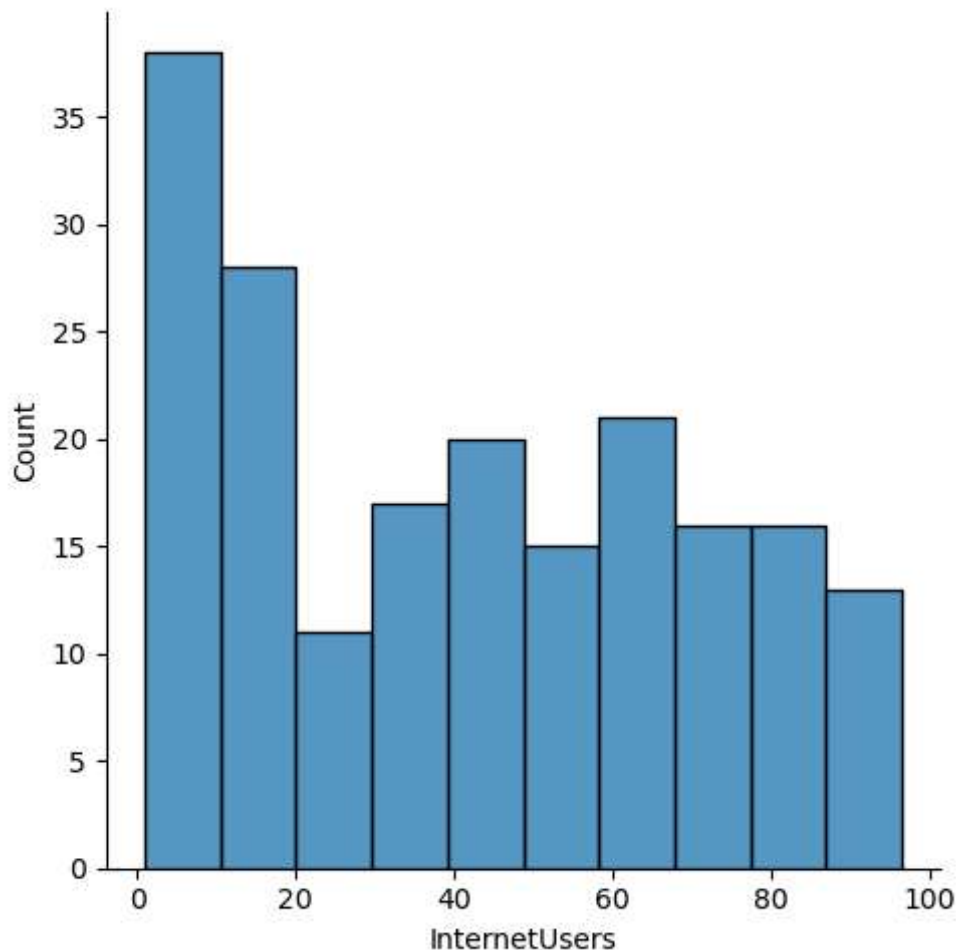
```
In [83]: vis2 = sns.boxplot(data = df, x="IncomeGroup", y='BirthRate') #biivariate  
vis2
```

```
Out[83]: <Axes: xlabel='IncomeGroup', ylabel='BirthRate'>
```



```
In [82]: vis = sns.displot(df['InternetUsers'], bins = 10) #univariate  
vis
```

```
Out[82]: <seaborn.axisgrid.FacetGrid at 0x173cf72e850>
```



1. Univariate analysis = plot the graph using 1 variable is called univariate analysis

ex: distplot,displot

2.Bivariate analysis =plot the graph using 2 variable is called bivariate analysis

ex :boxplot

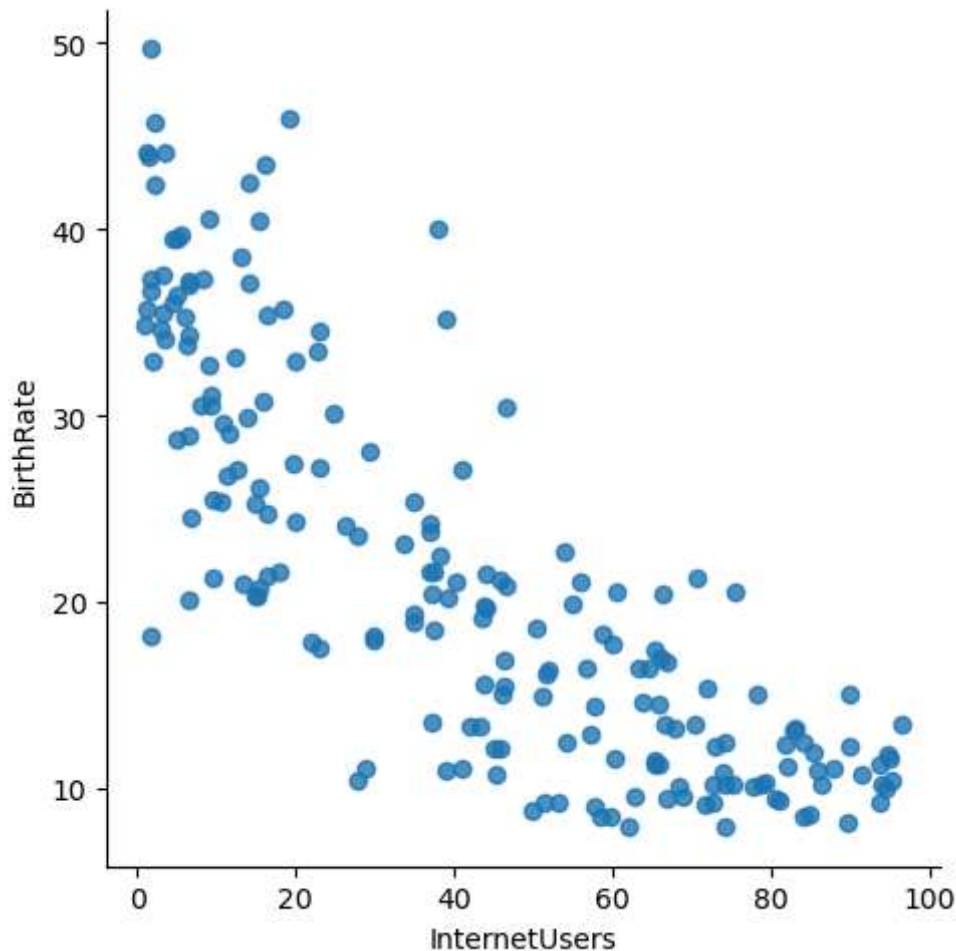
3.multivariate analysis = plot the graph using many variable is called bivariate analysis

outlier = anomaly detection

outlier = In stats ,outlier is the data point which is very far from other observation

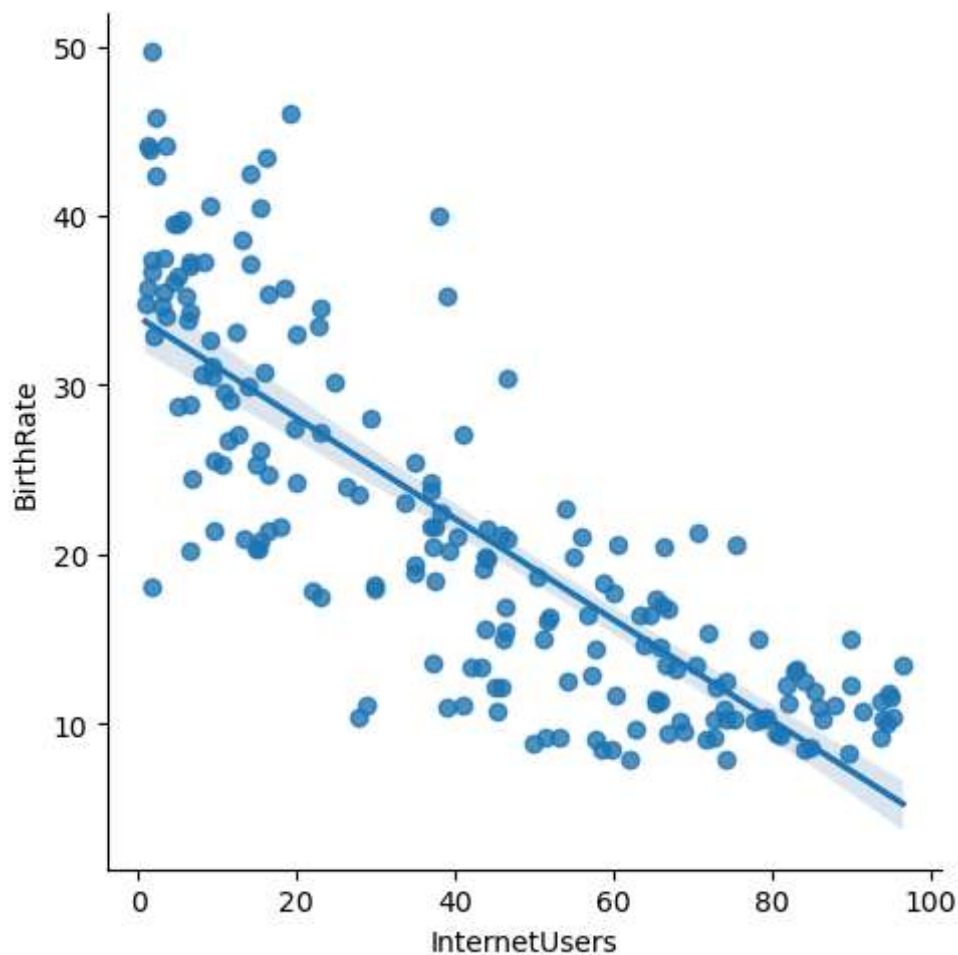
```
In [85]: vis3 = sns.lmplot(data = df,x = 'InternetUsers', y = 'BirthRate', fit_reg = False)
vis3
```

```
Out[85]: <seaborn.axisgrid.FacetGrid at 0x173cace0190>
```



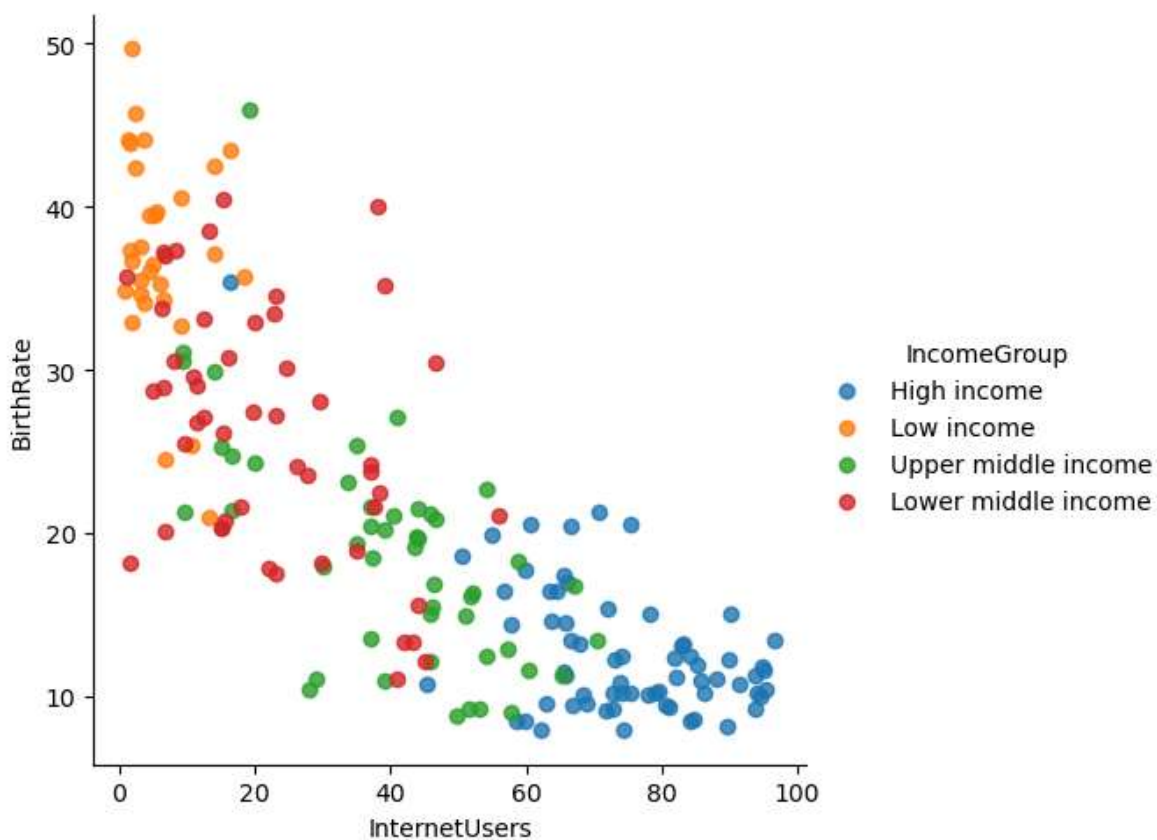
```
In [86]: vis3 = sns.lmplot(data = df,x = 'InternetUsers', y = 'BirthRate', fit_reg = True)
vis3
```

```
Out[86]: <seaborn.axisgrid.FacetGrid at 0x173ce251e50>
```



```
In [98]: vis3 = sns.lmplot(data = df,x = 'InternetUsers', y = 'BirthRate', fit_reg = False)
vis3
```

```
Out[98]: <seaborn.axisgrid.FacetGrid at 0x173d1325590>
```



```
In [ ]: 
```

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In [ ]: 
```

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In [ ]: 
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```
In [ ]: 
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